LAPORAN PRAKTIKUM PERANCANGAN JARINGAN

Pemilihan dan Penerapan Protokol Switching dan Routing #4



Disusun oleh

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PROGRAM STUDI DIPLOMA IV TEKNOLOGI REKAYASA INTERNET
DEPARTEMEN TEKNIK ELEKTRO DAN INFORMATIKA SEKOLAH VOKASI
UNIVERSITAS GADJAH MADA

2026

TUJUAN

a. Memahami perancangan jaringan secara top-down
b. Menerapkan rancangan jaringan berhirarki
c. Melakukan konfigurasi protokol-protokol
   - VLAN
   - Subnetting dan Pengalamatan IP
   - Inter-VLAN Router
   - DHCP
   - SSH
   - OSPF
   - Switching Security
   - Host
d. Pengujian dan Verifikasi komunikasi jaringan

DASAR TEORI

Faktor-faktor yang dilibatkan dalam sebuah pengambilan keputusan :

- Tujuan harus jelas
- Pilihan-pilihan atau ide yang ada harus dipelajari lebih jauh terlebih dahulu.
- Konsekuensi dari setiap keputusan harus dipelajari lebih lanjut
- Rencana cadangan harus segera dibuat.

Selain faktor-faktor diatas juga harus melibatkan beberapa pertanyaan berikut ini :

- Jika hasil keputusan yang diambil ternyata salah, apa yang akan dikerjakan ?
- Jika hasil keputusan sudah pernah dilakukan sebelumnya, apakah ada permasalahan dengan keputusan tersebut ?
- Bagaimana reaksi pelanggan dengan keputusan anda ?
- Apa rencana cadangan anda jika keputusan anda ditolak oleh pelanggan ?

Berikut ini adalah salah satu contoh dokumen dalam pembuatan keputusan

	Critical Goals			Other Goals		
	Adaptability—must adapt to changes in a large internetwork within seconds	Must scale to a large size (hundreds of routers)	Must be an industry standard and compatible with existing equipment	Should not create a lot of traffic	Should run on inexpensive routers	Should be easy to configure and manage
BGP	X*	X	X	8	7	7
OSPF	X	X	X	8	8	8
IS-IS	X	X	X	8	6	6
IGRP	X	X				
EIGRP	X	X				
RIP			X			

*X = Meets critical criteria. 1 = Lowest. 10 = Highest.

Pemilihan Switching Protocols

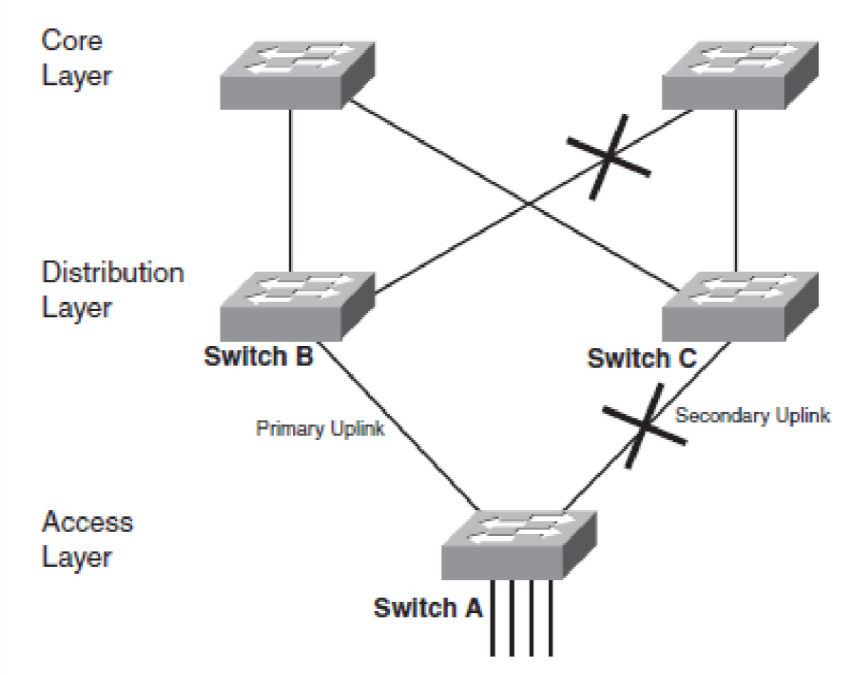
Switch memanfaatkan sirkuit terintegrasi cepat untuk menawarkan latensi rendah. Bridges jauh lebih lambat daripada switch dan memiliki lebih sedikit port dan biaya per port yang lebih tinggi. Untuk alasan ini, switch telah menggantikan bridges. Switch memiliki kemampuan untuk melakukan pemrosesan store-and-forward atau pemrosesan cut-through. Dengan pemrosesan cut-through, sebuah switch dengan cepat melihat alamat tujuan (bidang pertama dalam header frame Ethernet), menentukan port keluar, dan segera mulai mengirim bit ke port keluar. Kerugian dengan pemrosesan cut-through adalah meneruskan frame ilegal (misalnya, kerdil Ethernet) dan frame dengan kesalahan CRC.

Pada jaringan yang rawan kegagalan dan kesalahan, pemrosesan cut-through tidak boleh digunakan. Beberapa switch memiliki kemampuan untuk secara otomatis berpindah dari mode cut-through ke mode store-and forward ketika ambang kesalahan tercapai. Fitur ini disebut pengalihan cut-through adaptif oleh beberapa vendor. Switch memungkinkan beberapa jalur penerusan paralel, yang berarti sebuah switch dapat menangani volume lalu lintas yang tinggi lebih cepat daripada sebuah bridge. Switch high-end dapat mendukung banyak jalur penerusan simultan, tergantung pada struktur switching.

Berikut ini beberapa protokol switching yang dapat digunakan :

- STP
- RSTP
- PVST
- PVST+
- Etherchannel
- VTP
- DTP

Berikut ini contoh penggunaan STP



Gambar 1 Switch lapisan akses dengan dua uplinks ke switches layer distribusi

Pemilihan Routing Protocols

Protokol routing memungkinkan router secara dinamis mempelajari cara menjangkau jaringan lain dan bertukar informasi ini dengan router atau host lain. Memilih protokol routing untuk pelanggan desain jaringan Anda lebih sulit daripada memilih protokol switching, karena ada begitu banyak pilihan. Keputusan akan lebih mudah jika Anda dapat menggunakan tabel keputusan, seperti yang ditunjukkan pada Tabel 7-1 pada buku Top-Down Network Design 3rd Edition. Berbekal pemahaman yang kuat tentang tujuan dan informasi pelanggan Anda tentang karakteristik protokol routing yang berbeda, Anda dapat membuat keputusan yang tepat tentang protokol routing mana yang direkomendasikan. Berikut ini tabel ringkasan panduan dalam pemilihan protocol routing

Tabel 1 Perbandingan protokol-protokol routing

	Distance Vector or Link State	Interior or Exterior	Classful or Classless	Metrics Supported	Scalability	Convergence Time	Resource Consumption	Supports Security? Authenticates Routes?	Ease of Design, Configuration, and Troubleshooting
RIPv1	Distance vector	Interior	Classful	Hop count	15 hops	Can be long (if no load balancing)	Memory: low CPU: low Bandwidth: high	No	Easy
RIPv2	Distance vector	Interior	Classless	Hop count	15 hops	Can be long (if no load balancing)	Memory: low CPU: low Bandwidth: high	Yes	Easy
IGRP	Distance vector	Interior	Classful	Bandwidth, delay, reliability, load	255 hops (default is 100)	Quick (uses triggered updates and poison reverse)	Memory: low CPU: low Bandwidth: high	No	Easy
EIGRP	Advanced distance vector	Interior	Classless	Bandwidth, delay, reliability, load	1000s of routers	Very quick (uses DUAL algorithm)	Memory: moderate CPU: low Bandwidth: low	Yes	Easy
OSPF	Link state	Interior	Classless	Cost (100 million divided by bandwidth on Cisco routers)	A few hundred routers per area, a few hundred areas	Quick (uses LSAs and Hello packets)	Memory: high CPU: high Bandwidth: low	Yes	Moderate
BGP	Path vector	Exterior	Classless	Value of path attributes and other configurable factors	1000s of routers	Quick (uses update and keepalive packets, and withdraws routes)	Memory: high CPU: high Bandwidth: low	Yes	Moderate
IS-IS	Link state	Interior	Classless	Configured path value, plus delay, expense, and errors	Hundreds of routers per area, a few hundred areas	Quick (uses LSAs)	Memory: high CPU: high Bandwidth: low	Yes	Moderate

Berikut juga adalah tabel administrative distance berdasarkan tipe route untuk preferensi penentuan rute oleh router.

Tabel 2 Administrative Distance berdasarkan tipe route

Route Source	Default Distance Value
Connected interface	0
Static route	1
Enhanced Interior Gateway Routing Protocol (EIGRP) summary route	5
External Border Gateway Protocol (BGP)	20
Internal EIGRP	90
IGRP	100
OSPF	110
Intermediate System-to-Intermediate System (IS-IS)	115
Routing Information Protocol (RIP)	120
Exterior Gateway Protocol (EGP)	140
On-Demand Routing (ODR)	160
External EIGRP	170
Internal BGP	200
Unknown	255

Simpulan Pemilihan Protokol

Pemilihan protokol switching dan routing yang tepat untuk pelanggan desain jaringan Anda sangat diperlukan. Skalabilitas dan karakteristik kinerja protokol dan seberapa cepat protokol dapat beradaptasi dengan perubahan menjadi perhatian penting dalam pemilihan protocol witching maupun routing.

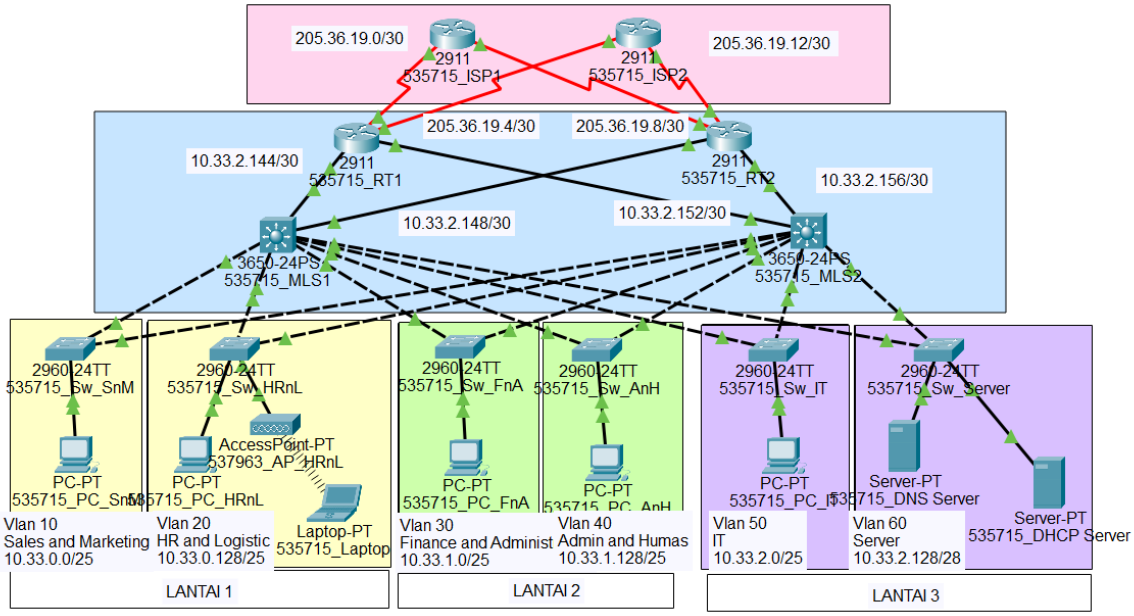
Memutuskan protokol switching dan routing yang tepat untuk pelanggan Anda akan membantu Anda memilih produk switch dan router terbaik untuk pelanggan. Misalnya, jika Anda telah memutuskan bahwa desain harus mendukung protokol routing yang dapat menyatu dalam hitungan detik di internetwork besar, Anda mungkin tidak akan merekomendasikan router yang hanya menjalankan RIP.

Detail pembahasan bagaimana memilih protocol switching dan routing sangaat direkomendasikan untuk membaca buku Top-down Network Design 3rd Edition dari CISCO yang juga sebagai buku utama matakuliah teori Perancangan Jaringan. Buku tersebut terdapat diskusi umum tentang pengambilan keputusan untuk membantu Anda mengembangkan proses sistematis untuk memilih solusi desain jaringan. Sebuah diskusi tentang menjembatani dan beralih protokol diikuti, yang meliputi menjembatani transparan, beralih multilayer, perangkat tambahan untuk protokol STP, dan VLAN. Bagian pada protokol routing mengikuti bagian switching.

PROSEDUR PERCOBAAN

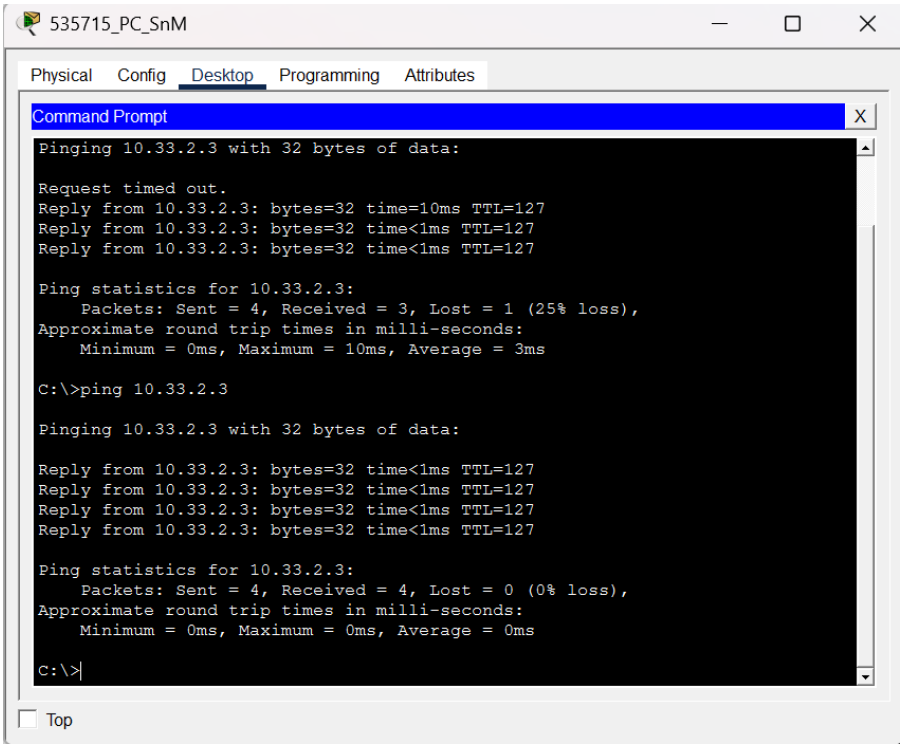
HASIL PRAKTIKUM

1. Hasil topologi (**lampirkan screenshot topologi dari packet tracer** masing-masing.



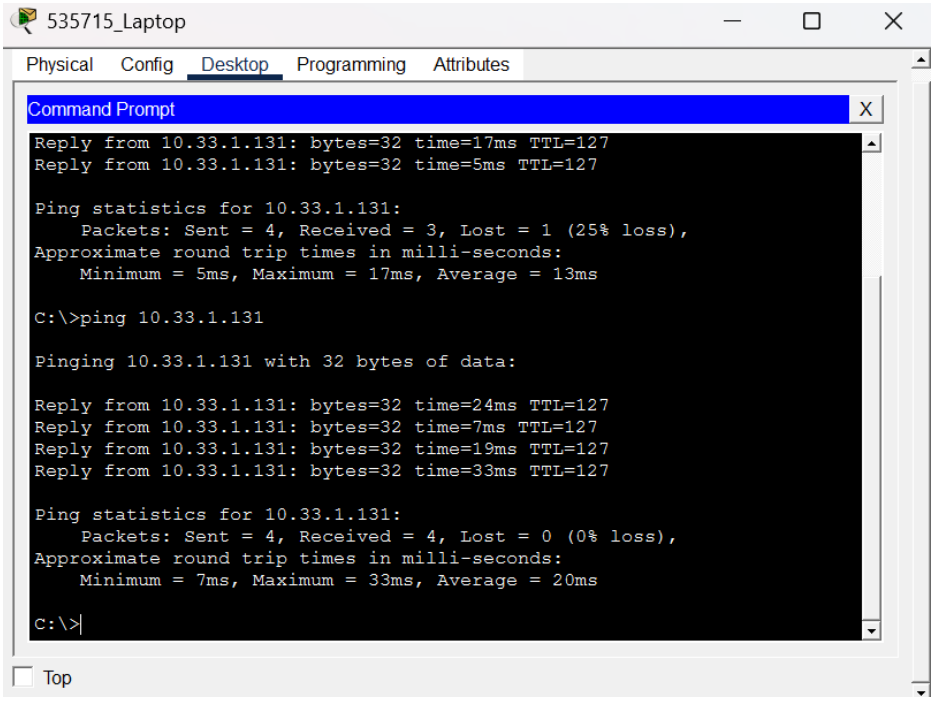
2. Uji koneksi - Ping dari salah satu komputer ke komputer lain **(jelaskan dari perangkat mana ke perangkat mana)**

   PC SnM ke PC IT



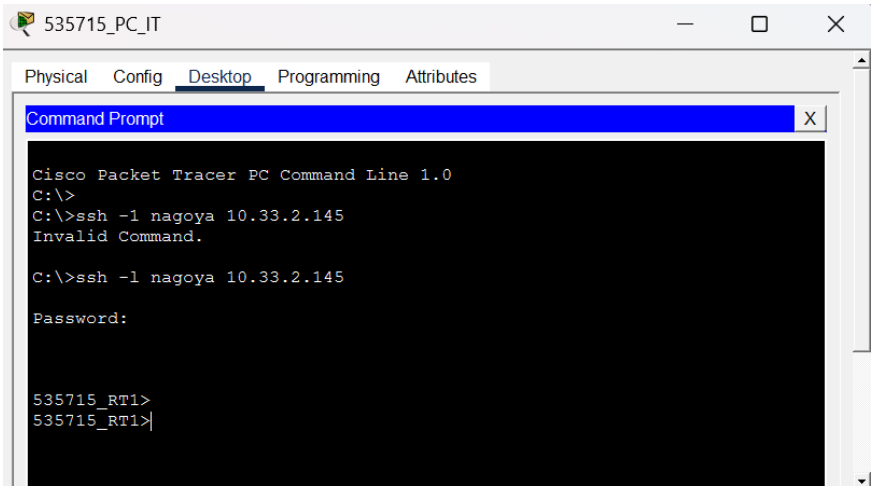
3. Uji koneksi wireless - Ping dari perangkat laptop ke salah satu perangkat lain.

   Laptop ke PC AnH



4. Uji SSH dari salah satu komputer ke salah satu perangkat router

   PC IT ke router 1



5. Verifikasi Interface

   ISP 1

```
Device Name: 535715_ISP1
Device Model: 2911
Hostname: 535715_ISP1

Port                Link   VLAN   IP Address         IPv6 Address                              MAC Address
GigabitEthernet0/0  Down   --     <not set>          <not set>                                 0060.7087.2601
GigabitEthernet0/1  Down   --     <not set>          <not set>                                 0060.7087.2602
GigabitEthernet0/2  Down   --     <not set>          <not set>                                 0060.7087.2603
Serial0/3/0         Up     --     205.36.19.1/30     <not set>                                 <not set>
Serial0/3/1         Up     --     205.36.19.9/30     <not set>                                 <not set>
Vlan1               Down   1      <not set>          <not set>                                 00E0.8FEE.3517

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715 ISP1
```

ISP 2

```
Device Name: 535715_ISP2
Device Model: 2911
Hostname: 535715_ISP2

Port                 Link   VLAN   IP Address         IPv6 Address                              MAC Address
GigabitEthernet0/0   Down   --     <not set>          <not set>                                 0001.6357.6801
GigabitEthernet0/1   Down   --     <not set>          <not set>                                 0001.6357.6802
GigabitEthernet0/2   Down   --     <not set>          <not set>                                 0001.6357.6803
Serial0/3/0          Up     --     205.36.19.5/30     <not set>                                 <not set>
Serial0/3/1          Up     --     205.36.19.13/30    <not set>                                 <not set>
Vlan1                Down   1      <not set>          <not set>                                 0040.0BE8.87DB

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715_ISP2
```

Router 1

```
Device Name: 535715_RT1
Device Model: 2911
Hostname: 535715_RT1

Port                 Link   VLAN   IP Address         IPv6 Address                              MAC Address
GigabitEthernet0/0   Up     --     10.33.2.145/30     <not set>                                 0005.5E38.6C01
GigabitEthernet0/1   Up     --     10.33.2.153/30     <not set>                                 0005.5E38.6C02
GigabitEthernet0/2   Down   --     <not set>          <not set>                                 0005.5E38.6C03
Serial0/3/0          Up     --     205.36.19.2/30     <not set>                                 <not set>
Serial0/3/1          Up     --     205.36.19.6/30     <not set>                                 <not set>
Vlan1                Down   1      <not set>          <not set>                                 0040.0B8E.9E77

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715_RT1
```

Router 2

```
Device Name: 535715_RT2
Device Model: 2911
Hostname: 535715_RT2

Port                 Link   VLAN   IP Address         IPv6 Address                              MAC Address
GigabitEthernet0/0   Up     --     10.33.2.157/30     <not set>                                 0050.0FE4.9401
GigabitEthernet0/1   Up     --     10.33.2.150/30     <not set>                                 0050.0FE4.9402
GigabitEthernet0/2   Down   --     <not set>          <not set>                                 0050.0FE4.9403
Serial0/3/0          Up     --     205.36.19.10/30    <not set>                                 <not set>
Serial0/3/1          Up     --     205.36.19.14/30    <not set>                                 <not set>
Vlan1                Down   1      <not set>          <not set>                                 0005.5E08.5B71

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715_RT2
```

MLS 1

```
Device Name: 535715_MLS1
Device Model: 3650-24PS
Hostname: 535715_MLS1

Port                      Link   VLAN   IP Address         IPv6 Address                              MAC Address
GigabitEthernet1/0/1      Up     1      10.33.2.146/30     <not set>                                 0001.64BC.DE01
GigabitEthernet1/0/2      Up     1      10.33.2.149/30     <not set>                                 0001.64BC.DE02
GigabitEthernet1/0/3      Up     --     <not set>          <not set>                                 0001.64BC.DE03
GigabitEthernet1/0/4      Up     --     <not set>          <not set>                                 0001.64BC.DE04
GigabitEthernet1/0/5      Up     --     <not set>          <not set>                                 0001.64BC.DE05
GigabitEthernet1/0/6      Up     --     <not set>          <not set>                                 0001.64BC.DE06
GigabitEthernet1/0/7      Up     --     <not set>          <not set>                                 0001.64BC.DE07
GigabitEthernet1/0/8      Up     --     <not set>          <not set>                                 0001.64BC.DE08
GigabitEthernet1/0/9      Down   1      <not set>          <not set>                                 0001.64BC.DE09
GigabitEthernet1/0/10     Down   1      <not set>          <not set>                                 0001.64BC.DE0A
GigabitEthernet1/0/11     Down   1      <not set>          <not set>                                 0001.64BC.DE0B
GigabitEthernet1/0/12     Down   1      <not set>          <not set>                                 0001.64BC.DE0C
GigabitEthernet1/0/13     Down   1      <not set>          <not set>                                 0001.64BC.DE0D
GigabitEthernet1/0/14     Down   1      <not set>          <not set>                                 0001.64BC.DE0E
GigabitEthernet1/0/15     Down   1      <not set>          <not set>                                 0001.64BC.DE0F
GigabitEthernet1/0/16     Down   1      <not set>          <not set>                                 0001.64BC.DE10
GigabitEthernet1/0/17     Down   1      <not set>          <not set>                                 0001.64BC.DE11
GigabitEthernet1/0/18     Down   1      <not set>          <not set>                                 0001.64BC.DE12
GigabitEthernet1/0/19     Down   1      <not set>          <not set>                                 0001.64BC.DE13
GigabitEthernet1/0/20     Down   1      <not set>          <not set>                                 0001.64BC.DE14
GigabitEthernet1/0/21     Down   1      <not set>          <not set>                                 0001.64BC.DE15
GigabitEthernet1/0/22     Down   1      <not set>          <not set>                                 0001.64BC.DE16
GigabitEthernet1/0/23     Down   1      <not set>          <not set>                                 0001.64BC.DE17
GigabitEthernet1/0/24     Down   1      <not set>          <not set>                                 0001.64BC.DE18
GigabitEthernet1/1/1      Down   1      <not set>          <not set>                                 000C.CFCD.9901
GigabitEthernet1/1/2      Down   1      <not set>          <not set>                                 000C.CFCD.9902
GigabitEthernet1/1/3      Down   1      <not set>          <not set>                                 000C.CFCD.9903
GigabitEthernet1/1/4      Down   1      <not set>          <not set>                                 000C.CFCD.9904
Vlan1                     Down   1      <not set>          <not set>                                 0002.1740.C521
Vlan10                    Up     10     10.33.0.1/25       <not set>                                 0002.1740.C501
Vlan20                    Up     20     10.33.0.129/25     <not set>                                 0002.1740.C502
Vlan30                    Up     30     10.33.1.1/25       <not set>                                 0002.1740.C503
Vlan40                    Up     40     10.33.1.129/25     <not set>                                 0002.1740.C504
Vlan50                    Up     50     10.33.2.1/25       <not set>                                 0002.1740.C505
Vlan60                    Up     60     10.33.2.129/28     <not set>                                 0002.1740.C506

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```

MLS 2

```
Device Name: 535715_MLS2
Device Model: 3650-24PS
Hostname: 535715_MLS2

Port                    Link   VLAN   IP Address         IPv6 Address                              MAC Address
GigabitEthernet1/0/1    Up     1      10.33.2.158/30     <not set>                                 0001.97AB.A001
GigabitEthernet1/0/2    Up     1      10.33.2.154/30     <not set>                                 0001.97AB.A002
GigabitEthernet1/0/3    Up     --     <not set>          <not set>                                 0001.97AB.A003
GigabitEthernet1/0/4    Up     --     <not set>          <not set>                                 0001.97AB.A004
GigabitEthernet1/0/5    Up     --     <not set>          <not set>                                 0001.97AB.A005
GigabitEthernet1/0/6    Up     --     <not set>          <not set>                                 0001.97AB.A006
GigabitEthernet1/0/7    Up     --     <not set>          <not set>                                 0001.97AB.A007
GigabitEthernet1/0/8    Up     --     <not set>          <not set>                                 0001.97AB.A008
GigabitEthernet1/0/9    Down   1      <not set>          <not set>                                 0001.97AB.A009
GigabitEthernet1/0/10   Down   1      <not set>          <not set>                                 0001.97AB.A00A
GigabitEthernet1/0/11   Down   1      <not set>          <not set>                                 0001.97AB.A00B
GigabitEthernet1/0/12   Down   1      <not set>          <not set>                                 0001.97AB.A00C
GigabitEthernet1/0/13   Down   1      <not set>          <not set>                                 0001.97AB.A00D
GigabitEthernet1/0/14   Down   1      <not set>          <not set>                                 0001.97AB.A00E
GigabitEthernet1/0/15   Down   1      <not set>          <not set>                                 0001.97AB.A00F
GigabitEthernet1/0/16   Down   1      <not set>          <not set>                                 0001.97AB.A010
GigabitEthernet1/0/17   Down   1      <not set>          <not set>                                 0001.97AB.A011
GigabitEthernet1/0/18   Down   1      <not set>          <not set>                                 0001.97AB.A012
GigabitEthernet1/0/19   Down   1      <not set>          <not set>                                 0001.97AB.A013
GigabitEthernet1/0/20   Down   1      <not set>          <not set>                                 0001.97AB.A014
GigabitEthernet1/0/21   Down   1      <not set>          <not set>                                 0001.97AB.A015
GigabitEthernet1/0/22   Down   1      <not set>          <not set>                                 0001.97AB.A016
GigabitEthernet1/0/23   Down   1      <not set>          <not set>                                 0001.97AB.A017
GigabitEthernet1/0/24   Down   1      <not set>          <not set>                                 0001.97AB.A018
GigabitEthernet1/1/1    Down   1      <not set>          <not set>                                 0060.3E68.E301
GigabitEthernet1/1/2    Down   1      <not set>          <not set>                                 0060.3E68.E302
GigabitEthernet1/1/3    Down   1      <not set>          <not set>                                 0060.3E68.E303
GigabitEthernet1/1/4    Down   1      <not set>          <not set>                                 0060.3E68.E304
Vlan1                   Down   1      <not set>          <not set>                                 000A.4179.D71A
Vlan10                  Up     10     10.33.0.2/25       <not set>                                 000A.4179.D701
Vlan20                  Up     20     10.33.0.130/25     <not set>                                 000A.4179.D702
Vlan30                  Up     30     10.33.1.2/25       <not set>                                 000A.4179.D703
Vlan40                  Up     40     10.33.1.130/25     <not set>                                 000A.4179.D704
Vlan50                  Up     50     10.33.2.2/25       <not set>                                 000A.4179.D705
Vlan60                  Up     60     10.33.2.130/28     <not set>                                 000A.4179.D706

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715_MLS2
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Switch Sales and Marketing

```
Device Name: 535715_Sw_SnM
Custom Device Model: 2960 IOS15
Hostname: 5355715_Sw_SnM

Port                Link   VLAN   IP Address         MAC Address
FastEthernet0/1     Up     --     --                 0001.96D7.9A01
FastEthernet0/2     Up     --     --                 0001.96D7.9A02
FastEthernet0/3     Up     10     --                 0001.96D7.9A03
FastEthernet0/4     Down   10     --                 0001.96D7.9A04
FastEthernet0/5     Down   10     --                 0001.96D7.9A05
FastEthernet0/6     Down   10     --                 0001.96D7.9A06
FastEthernet0/7     Down   10     --                 0001.96D7.9A07
FastEthernet0/8     Down   10     --                 0001.96D7.9A08
FastEthernet0/9     Down   10     --                 0001.96D7.9A09
FastEthernet0/10    Down   10     --                 0001.96D7.9A0A
FastEthernet0/11    Down   10     --                 0001.96D7.9A0B
FastEthernet0/12    Down   10     --                 0001.96D7.9A0C
FastEthernet0/13    Down   10     --                 0001.96D7.9A0D
FastEthernet0/14    Down   10     --                 0001.96D7.9A0E
FastEthernet0/15    Down   10     --                 0001.96D7.9A0F
FastEthernet0/16    Down   10     --                 0001.96D7.9A10
FastEthernet0/17    Down   10     --                 0001.96D7.9A11
FastEthernet0/18    Down   10     --                 0001.96D7.9A12
FastEthernet0/19    Down   10     --                 0001.96D7.9A13
FastEthernet0/20    Down   10     --                 0001.96D7.9A14
FastEthernet0/21    Down   10     --                 0001.96D7.9A15
FastEthernet0/22    Down   10     --                 0001.96D7.9A16
FastEthernet0/23    Down   10     --                 0001.96D7.9A17
FastEthernet0/24    Down   10     --                 0001.96D7.9A18
GigabitEthernet0/1  Down   1      --                 0001.96D7.9A19
GigabitEthernet0/2  Down   1      --                 0001.96D7.9A1A
Vlan1               Down   1      <not set>          0005.5EEB.9E01

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715_Sw_SnM
```

Switch HR and Logistic

```
Device Name: 535715_Sw_HRnL
Custom Device Model: 2960 IOS15
Hostname: 535715_Sw_HRnL

Port                Link   VLAN   IP Address         MAC Address
FastEthernet0/1     Up     --     --                 0006.2A2D.DD01
FastEthernet0/2     Up     --     --                 0006.2A2D.DD02
FastEthernet0/3     Up     20     --                 0006.2A2D.DD03
FastEthernet0/4     Up     20     --                 0006.2A2D.DD04
FastEthernet0/5     Down   20     --                 0006.2A2D.DD05
FastEthernet0/6     Down   20     --                 0006.2A2D.DD06
FastEthernet0/7     Down   20     --                 0006.2A2D.DD07
FastEthernet0/8     Down   20     --                 0006.2A2D.DD08
FastEthernet0/9     Down   20     --                 0006.2A2D.DD09
FastEthernet0/10    Down   20     --                 0006.2A2D.DD0A
FastEthernet0/11    Down   20     --                 0006.2A2D.DD0B
FastEthernet0/12    Down   20     --                 0006.2A2D.DD0C
FastEthernet0/13    Down   20     --                 0006.2A2D.DD0D
FastEthernet0/14    Down   20     --                 0006.2A2D.DD0E
FastEthernet0/15    Down   20     --                 0006.2A2D.DD0F
FastEthernet0/16    Down   20     --                 0006.2A2D.DD10
FastEthernet0/17    Down   20     --                 0006.2A2D.DD11
FastEthernet0/18    Down   20     --                 0006.2A2D.DD12
FastEthernet0/19    Down   20     --                 0006.2A2D.DD13
FastEthernet0/20    Down   20     --                 0006.2A2D.DD14
FastEthernet0/21    Down   20     --                 0006.2A2D.DD15
FastEthernet0/22    Down   20     --                 0006.2A2D.DD16
FastEthernet0/23    Down   20     --                 0006.2A2D.DD17
FastEthernet0/24    Down   20     --                 0006.2A2D.DD18
GigabitEthernet0/1  Down   1      --                 0006.2A2D.DD19
GigabitEthernet0/2  Down   1      --                 0006.2A2D.DD1A
Vlan1               Down   1      <not set>          0090.2BDD.C108

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715_Sw_HRnL
```

Switch Finance and Administration

```
Device Name: 535715_Sw_FnA
Custom Device Model: 2960 IOS15
Hostname: 535715_Sw_FnA

Port                Link   VLAN   IP Address         MAC Address
FastEthernet0/1     Up     --     --                 0001.968B.1101
FastEthernet0/2     Up     --     --                 0001.968B.1102
FastEthernet0/3     Up     30     --                 0001.968B.1103
FastEthernet0/4     Down   30     --                 0001.968B.1104
FastEthernet0/5     Down   30     --                 0001.968B.1105
FastEthernet0/6     Down   30     --                 0001.968B.1106
FastEthernet0/7     Down   30     --                 0001.968B.1107
FastEthernet0/8     Down   30     --                 0001.968B.1108
FastEthernet0/9     Down   30     --                 0001.968B.1109
FastEthernet0/10    Down   30     --                 0001.968B.110A
FastEthernet0/11    Down   30     --                 0001.968B.110B
FastEthernet0/12    Down   30     --                 0001.968B.110C
FastEthernet0/13    Down   30     --                 0001.968B.110D
FastEthernet0/14    Down   30     --                 0001.968B.110E
FastEthernet0/15    Down   30     --                 0001.968B.110F
FastEthernet0/16    Down   30     --                 0001.968B.1110
FastEthernet0/17    Down   30     --                 0001.968B.1111
FastEthernet0/18    Down   30     --                 0001.968B.1112
FastEthernet0/19    Down   30     --                 0001.968B.1113
FastEthernet0/20    Down   30     --                 0001.968B.1114
FastEthernet0/21    Down   30     --                 0001.968B.1115
FastEthernet0/22    Down   30     --                 0001.968B.1116
FastEthernet0/23    Down   30     --                 0001.968B.1117
FastEthernet0/24    Down   30     --                 0001.968B.1118
GigabitEthernet0/1  Down   1      --                 0001.968B.1119
GigabitEthernet0/2  Down   1      --                 0001.968B.111A
Vlan1               Down   1      <not set>          0007.EC71.B479

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715_Sw_FnA
```

Switch Admin and Humas

```
Device Name: 535715_Sw_AnH
Custom Device Model: 2960 IOS15
Hostname: 535715_Sw_AnH

Port                Link   VLAN   IP Address          MAC Address
FastEthernet0/1     Up     --     --                  000C.8529.E201
FastEthernet0/2     Up     --     --                  000C.8529.E202
FastEthernet0/3     Up     40     --                  000C.8529.E203
FastEthernet0/4     Down   40     --                  000C.8529.E204
FastEthernet0/5     Down   40     --                  000C.8529.E205
FastEthernet0/6     Down   40     --                  000C.8529.E206
FastEthernet0/7     Down   40     --                  000C.8529.E207
FastEthernet0/8     Down   40     --                  000C.8529.E208
FastEthernet0/9     Down   40     --                  000C.8529.E209
FastEthernet0/10    Down   40     --                  000C.8529.E20A
FastEthernet0/11    Down   40     --                  000C.8529.E20B
FastEthernet0/12    Down   40     --                  000C.8529.E20C
FastEthernet0/13    Down   40     --                  000C.8529.E20D
FastEthernet0/14    Down   40     --                  000C.8529.E20E
FastEthernet0/15    Down   40     --                  000C.8529.E20F
FastEthernet0/16    Down   40     --                  000C.8529.E210
FastEthernet0/17    Down   40     --                  000C.8529.E211
FastEthernet0/18    Down   40     --                  000C.8529.E212
FastEthernet0/19    Down   40     --                  000C.8529.E213
FastEthernet0/20    Down   40     --                  000C.8529.E214
FastEthernet0/21    Down   40     --                  000C.8529.E215
FastEthernet0/22    Down   40     --                  000C.8529.E216
FastEthernet0/23    Down   40     --                  000C.8529.E217
FastEthernet0/24    Down   40     --                  000C.8529.E218
GigabitEthernet0/1  Down   1      --                  000C.8529.E219
GigabitEthernet0/2  Down   1      --                  000C.8529.E21A
Vlan1               Down   1      <not set>           00E0.F7B4.66B7

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715 Sw AnH
```

Switch IT

```
Device Name: 535715_Sw_IT
Custom Device Model: 2960 IOS15
Hostname: 535715_Sw_IT

Port                Link   VLAN   IP Address          MAC Address
FastEthernet0/1     Up     --     --                  0001.63CD.0C01
FastEthernet0/2     Up     --     --                  0001.63CD.0C02
FastEthernet0/3     Up     50     --                  0001.63CD.0C03
FastEthernet0/4     Down   50     --                  0001.63CD.0C04
FastEthernet0/5     Down   50     --                  0001.63CD.0C05
FastEthernet0/6     Down   50     --                  0001.63CD.0C06
FastEthernet0/7     Down   50     --                  0001.63CD.0C07
FastEthernet0/8     Down   50     --                  0001.63CD.0C08
FastEthernet0/9     Down   50     --                  0001.63CD.0C09
FastEthernet0/10    Down   50     --                  0001.63CD.0C0A
FastEthernet0/11    Down   50     --                  0001.63CD.0C0B
FastEthernet0/12    Down   50     --                  0001.63CD.0C0C
FastEthernet0/13    Down   50     --                  0001.63CD.0C0D
FastEthernet0/14    Down   50     --                  0001.63CD.0C0E
FastEthernet0/15    Down   50     --                  0001.63CD.0C0F
FastEthernet0/16    Down   50     --                  0001.63CD.0C10
FastEthernet0/17    Down   50     --                  0001.63CD.0C11
FastEthernet0/18    Down   50     --                  0001.63CD.0C12
FastEthernet0/19    Down   50     --                  0001.63CD.0C13
FastEthernet0/20    Down   50     --                  0001.63CD.0C14
FastEthernet0/21    Down   50     --                  0001.63CD.0C15
FastEthernet0/22    Down   50     --                  0001.63CD.0C16
FastEthernet0/23    Down   50     --                  0001.63CD.0C17
FastEthernet0/24    Down   50     --                  0001.63CD.0C18
GigabitEthernet0/1  Down   1      --                  0001.63CD.0C19
GigabitEthernet0/2  Down   1      --                  0001.63CD.0C1A
Vlan1               Down   1      <not set>           0060.4754.5EAC

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715_Sw_IT
```

Switch Server

```
Device Name: 535715_Sw_Server
Custom Device Model: 2960 IOS15
Hostname: 535715_Sw_Server

Port                Link   VLAN   IP Address         MAC Address
FastEthernet0/1     Up     --     --                 0090.2BD3.7C01
FastEthernet0/2     Up     --     --                 0090.2BD3.7C02
FastEthernet0/3     Up     60     --                 0090.2BD3.7C03
FastEthernet0/4     Up     60     --                 0090.2BD3.7C04
FastEthernet0/5     Down   60     --                 0090.2BD3.7C05
FastEthernet0/6     Down   60     --                 0090.2BD3.7C06
FastEthernet0/7     Down   60     --                 0090.2BD3.7C07
FastEthernet0/8     Down   60     --                 0090.2BD3.7C08
FastEthernet0/9     Down   60     --                 0090.2BD3.7C09
FastEthernet0/10    Down   60     --                 0090.2BD3.7C0A
FastEthernet0/11    Down   60     --                 0090.2BD3.7C0B
FastEthernet0/12    Down   60     --                 0090.2BD3.7C0C
FastEthernet0/13    Down   60     --                 0090.2BD3.7C0D
FastEthernet0/14    Down   60     --                 0090.2BD3.7C0E
FastEthernet0/15    Down   60     --                 0090.2BD3.7C0F
FastEthernet0/16    Down   60     --                 0090.2BD3.7C10
FastEthernet0/17    Down   60     --                 0090.2BD3.7C11
FastEthernet0/18    Down   60     --                 0090.2BD3.7C12
FastEthernet0/19    Down   60     --                 0090.2BD3.7C13
FastEthernet0/20    Down   60     --                 0090.2BD3.7C14
FastEthernet0/21    Down   60     --                 0090.2BD3.7C15
FastEthernet0/22    Down   60     --                 0090.2BD3.7C16
FastEthernet0/23    Down   60     --                 0090.2BD3.7C17
FastEthernet0/24    Down   60     --                 0090.2BD3.7C18
GigabitEthernet0/1  Down   1      --                 0090.2BD3.7C19
GigabitEthernet0/2  Down   1      --                 0090.2BD3.7C1A
Vlan1               Down   1      <not set>          0060.7020.BBBC

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715_Sw_Server
```

PC Sales and Marketing

```
Device Name: 535715_PC_SnM
Device Model: PC-PT

Port            Link   IP Address        IPv6 Address                              MAC Address
FastEthernet0   Up     10.33.0.3/25      <not set>                                 0001.43E7.05AE
Bluetooth       Down   <not set>         <not set>                                 00E0.F7A1.354E

Gateway: 10.33.0.1
DNS Server: 10.33.2.131
Line Number:  <not set>

Physical Location: Intercity > Home City > Corporate Office > 535715_PC_SnM
```

PC HR and Logistic

```
Device Name: 537963_PC_HRnL
Device Model: PC-PT

Port            Link   IP Address        IPv6 Address                              MAC Address
FastEthernet0   Up     10.33.0.131/25    <not set>                                 0007.EC20.DA66
Bluetooth       Down   <not set>         <not set>                                 0004.9AC7.1310

Gateway: 10.33.0.129
DNS Server: 10.33.2.121
Line Number:  <not set>

Physical Location: Intercity > Home City > Corporate Office > 537963_PC_HRnL
```

PC Finance and Administration

```
Device Name: 537963_PC_FnA
Device Model: PC-PT

Port            Link   IP Address        IPv6 Address                              MAC Address
FastEthernet0   Up     10.33.1.3/25      <not set>                                 000C.CF4C.602B
Bluetooth       Down   <not set>         <not set>                                 0002.170E.3670

Gateway: 10.33.1.1
DNS Server: 10.33.2.131
Line Number:  <not set>

Physical Location: Intercity > Home City > Corporate Office > 537963_PC_FnA
```

PC Admin and Humas

```
Device Name: 535715_PC_AnH
Device Model: PC-PT

Port            Link   IP Address         IPv6 Address                              MAC Address
FastEthernet0   Up     10.33.1.131/28     <not set>                                 0030.F2A0.A77A
Bluetooth       Down   <not set>          <not set>                                 0002.4A61.2B7E

Gateway: 0.0.0.0
DNS Server: 10.33.2.131
Line Number:  <not set>

Physical Location: Intercity > Home City > Corporate Office > 535715 PC AnH
```

PC IT

```
Device Name: 535715_PC_IT
Device Model: PC-PT

Port            Link   IP Address         IPv6 Address                              MAC Address
FastEthernet0   Up     10.33.2.3/25       <not set>                                 0090.2BB4.C4EA
Bluetooth       Down   <not set>          <not set>                                 0060.4703.8580

Gateway: 10.33.2.1
DNS Server: 10.33.2.131
Line Number:  <not set>

Physical Location: Intercity > Home City > Corporate Office > 535715_PC_IT
```

Server DNS

```
Device Name: 535715_DNS Server
Device Model: Server-PT

Port            Link   IP Address         IPv6 Address                              MAC Address
FastEthernet0   Up     10.33.2.131/28     <not set>                                 00D0.BC55.13E6

Gateway: 10.33.2.129
DNS Server: 10.33.2.131
Line Number:  <not set>

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715_DNS Server
```

Server DHCP

```
Device Name: 535715_DHCP Server
Device Model: Server-PT

Port            Link   IP Address         IPv6 Address                              MAC Address
FastEthernet0   Up     10.33.2.133/28     <not set>                                 00D0.58D4.5624

Gateway: 10.33.2.129
DNS Server: 10.33.2.131
Line Number:  <not set>

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 535715_DHCP Server
```

Access Point

```
Device Name: 537963_AP_HRnL
Device Model: AccessPoint-PT

Port      Link
Port 0    Up
Port 1    Up

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > 537963_AP_HRnL
```

Laptop

```
Device Name: 535715_Laptop
Device Model: Laptop-PT

Port            Link   IP Address         IPv6 Address                              MAC Address
Wireless0       Up     10.33.0.131/28     <not set>                                 0030.A398.B0BB
Bluetooth       Down   <not set>          <not set>                                 0060.2FE8.019B

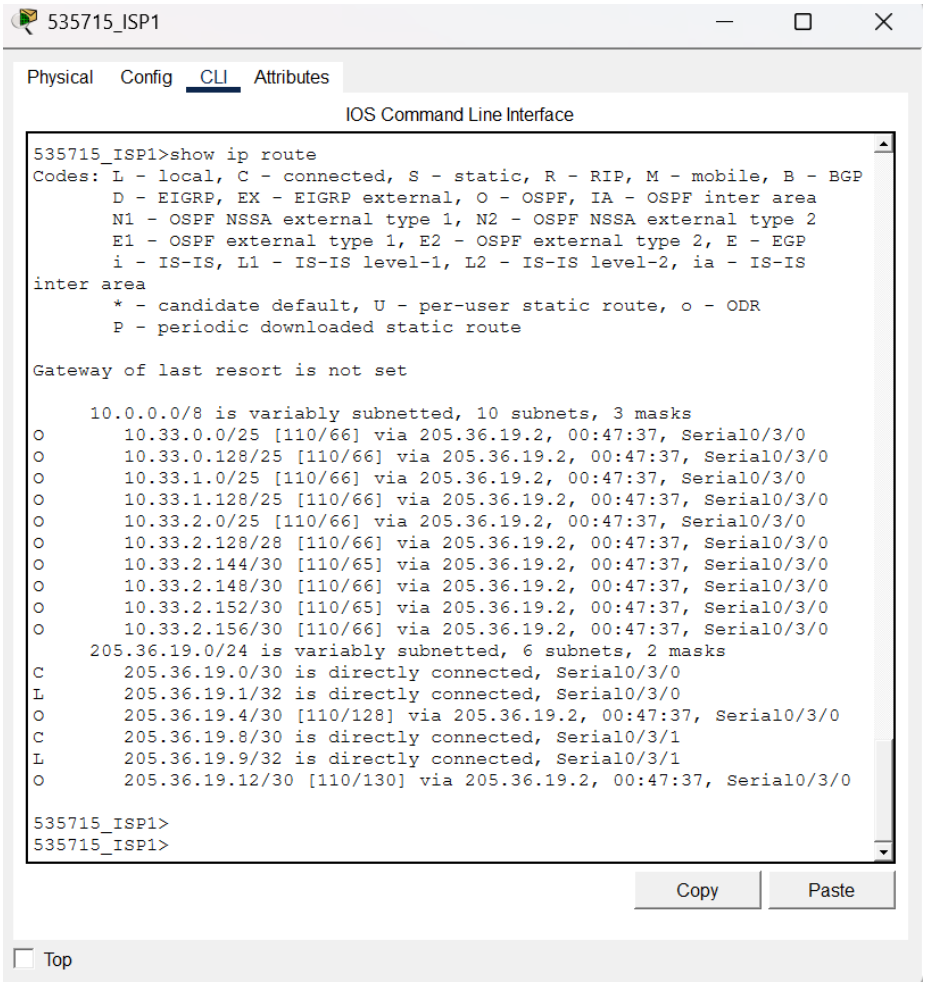
Gateway: 0.0.0.0
DNS Server: 10.33.2.131
Line Number:  <not set>

Wireless Best Data Rate: 54 Mbps
Wireless Signal Strength: 67%

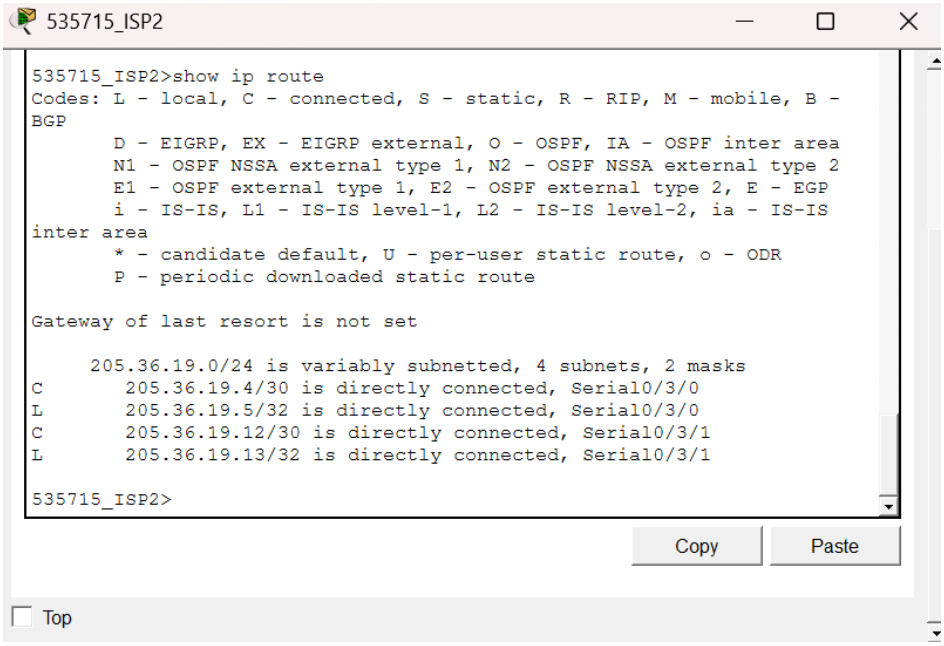
Physical Location: Intercity > Home City > Corporate Office > 535715 Laptop
```

6. Verifikasi routing

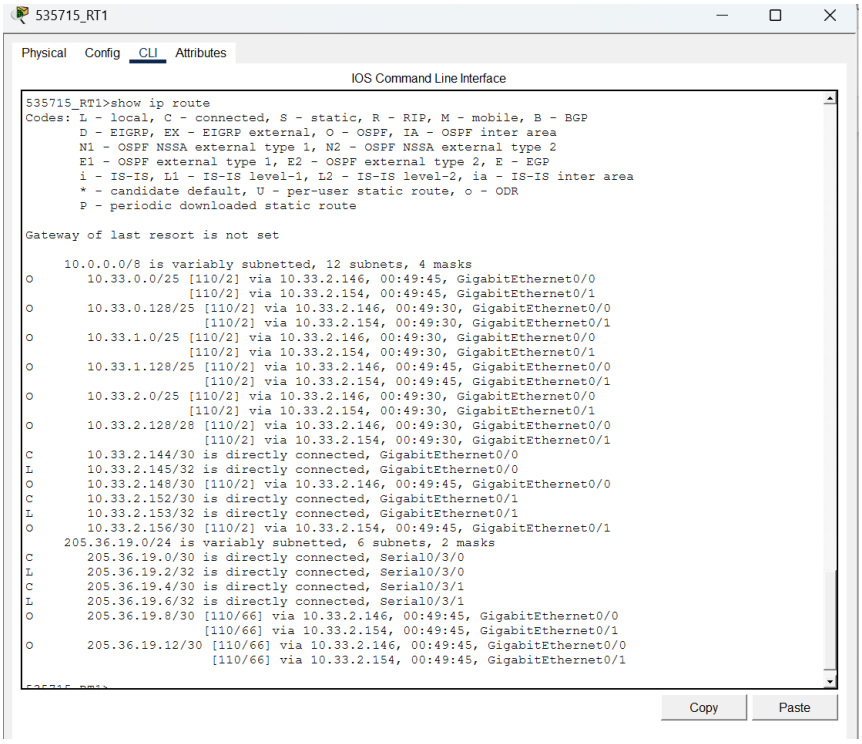
ISP 1



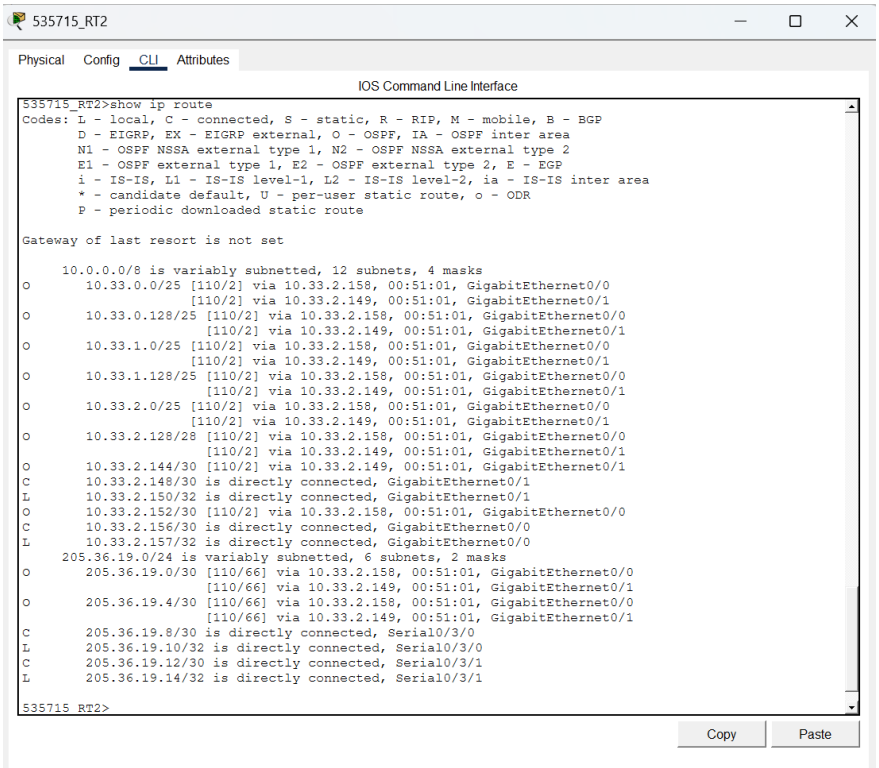
ISP 2



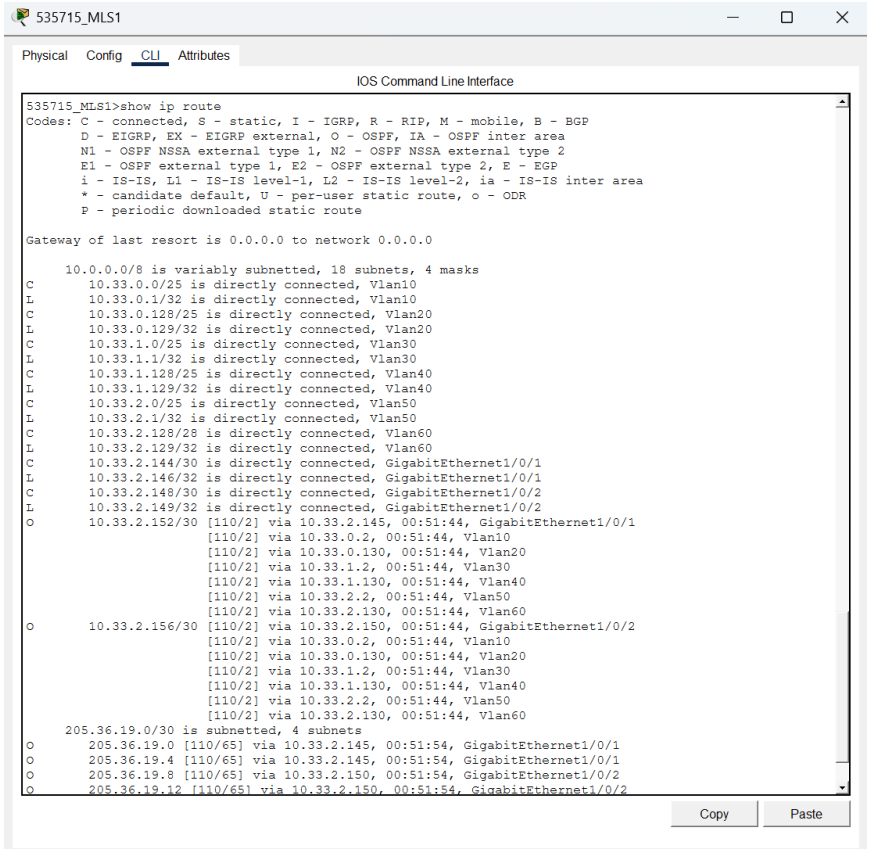
Router 1



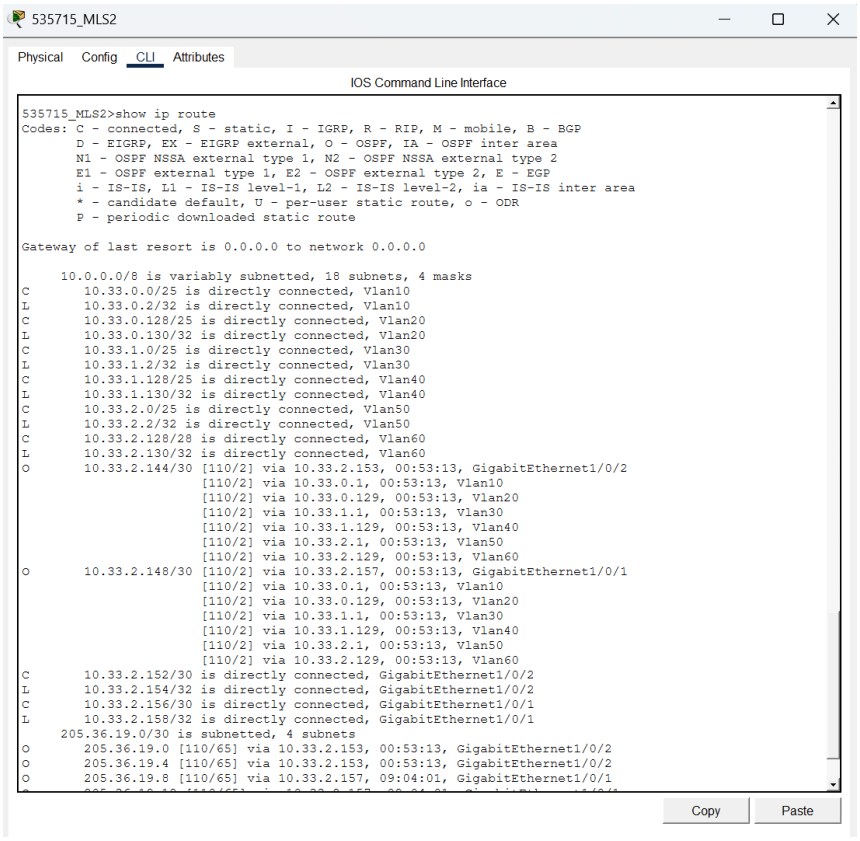
Router 2



MLS 1

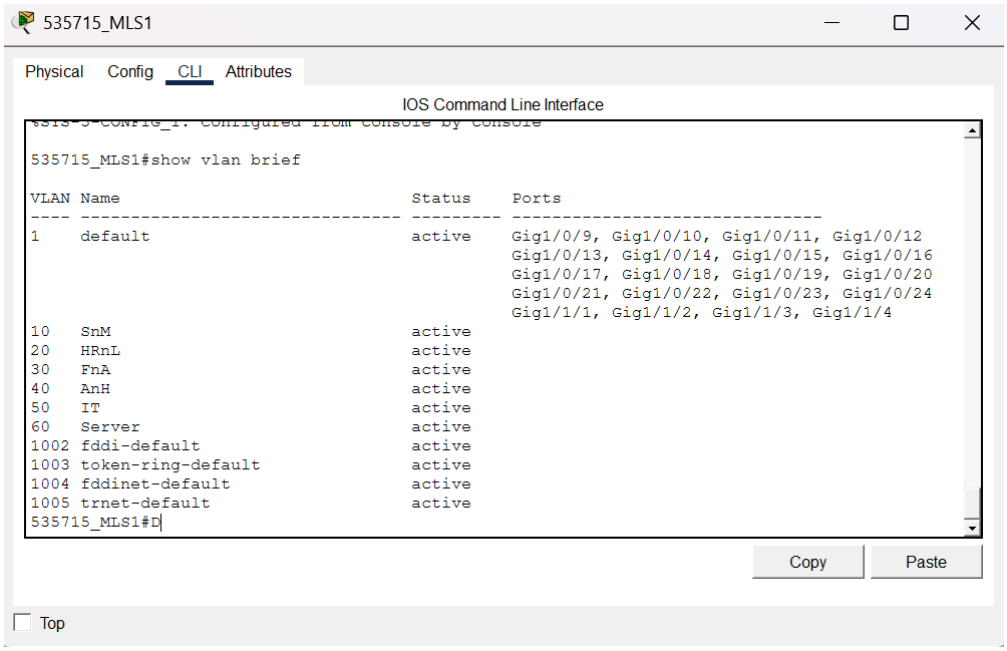

535715_MLS1
Physical
Config
CLI
Attributes
IOS Command Line Interface
535715_MLS1>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
10.0.0.0/8 is variably subnetted, 18 subnets, 4 masks
C 10.33.0.0/25 is directly connected, Vlan10
L 10.33.0.1/32 is directly connected, Vlan10
C 10.33.0.128/25 is directly connected, Vlan20
L 10.33.0.129/32 is directly connected, Vlan20
C 10.33.1.0/25 is directly connected, Vlan30
L 10.33.1.1/32 is directly connected, Vlan30
C 10.33.1.128/25 is directly connected, Vlan40
L 10.33.1.129/32 is directly connected, Vlan40
C 10.33.2.0/25 is directly connected, Vlan50
L 10.33.2.1/32 is directly connected, Vlan50
C 10.33.2.128/28 is directly connected, Vlan60
L 10.33.2.129/32 is directly connected, Vlan60
C 10.33.2.144/30 is directly connected, GigabitEthernet1/0/1
L 10.33.2.146/32 is directly connected, GigabitEthernet1/0/1
C 10.33.2.148/30 is directly connected, GigabitEthernet1/0/2
L 10.33.2.149/32 is directly connected, GigabitEthernet1/0/2
O 10.33.2.152/30 [110/2] via 10.33.2.145, 00:51:44, GigabitEthernet1/0/1
[110/2] via 10.33.0.2, 00:51:44, Vlan10
[110/2] via 10.33.0.130, 00:51:44, Vlan20
[110/2] via 10.33.1.2, 00:51:44, Vlan30
[110/2] via 10.33.1.130, 00:51:44, Vlan40
[110/2] via 10.33.2.2, 00:51:44, Vlan50
[110/2] via 10.33.2.130, 00:51:44, Vlan60
O 10.33.2.156/30 [110/2] via 10.33.2.150, 00:51:44, GigabitEthernet1/0/2
[110/2] via 10.33.0.2, 00:51:44, Vlan10
[110/2] via 10.33.0.130, 00:51:44, Vlan20
[110/2] via 10.33.1.2, 00:51:44, Vlan30
[110/2] via 10.33.1.130, 00:51:44, Vlan40
[110/2] via 10.33.2.2, 00:51:44, Vlan50
[110/2] via 10.33.2.130, 00:51:44, Vlan60
205.36.19.0/30 is subnetted, 4 subnets
O 205.36.19.0 [110/65] via 10.33.2.145, 00:51:54, GigabitEthernet1/0/1
O 205.36.19.4 [110/65] via 10.33.2.145, 00:51:54, GigabitEthernet1/0/1
O 205.36.19.8 [110/65] via 10.33.2.150, 00:51:54, GigabitEthernet1/0/2
O 205.36.19.12 [110/65] via 10.33.2.150, 00:51:54, GigabitEthernet1/0/2
Copy
Paste

MLS 2

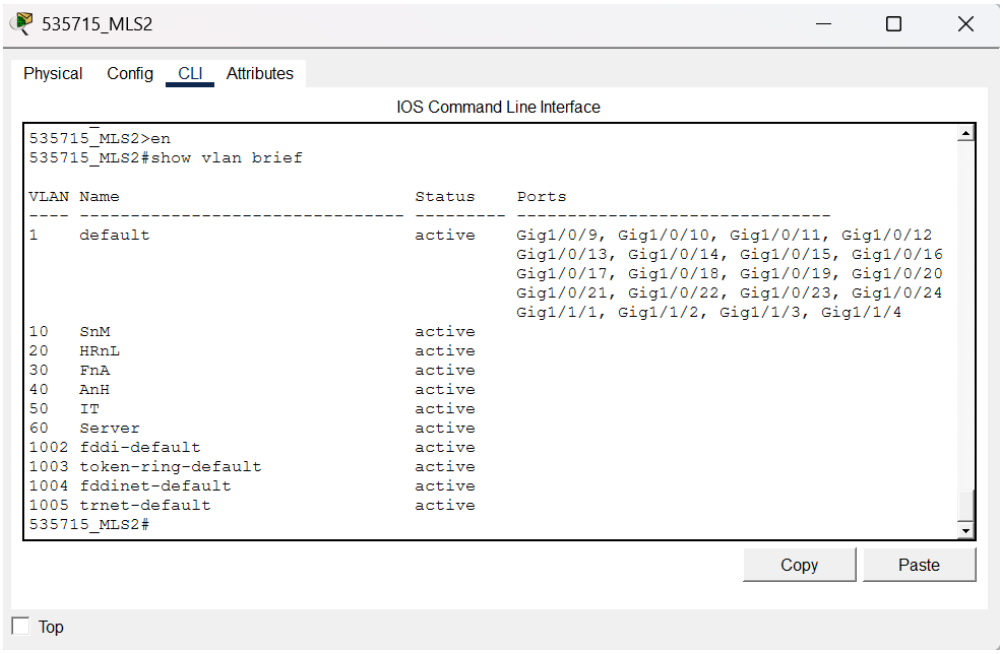

535715_MLS2
Physical
Config
CLI
Attributes
IOS Command Line Interface
535715_MLS2>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
10.0.0.0/8 is variably subnetted, 18 subnets, 4 masks
C 10.33.0.0/25 is directly connected, Vlan10
L 10.33.0.2/32 is directly connected, Vlan10
C 10.33.0.128/25 is directly connected, Vlan20
L 10.33.0.130/32 is directly connected, Vlan20
C 10.33.1.0/25 is directly connected, Vlan30
L 10.33.1.2/32 is directly connected, Vlan30
C 10.33.1.128/25 is directly connected, Vlan40
L 10.33.1.130/32 is directly connected, Vlan40
C 10.33.2.0/25 is directly connected, Vlan50
L 10.33.2.2/32 is directly connected, Vlan50
C 10.33.2.128/28 is directly connected, Vlan60
L 10.33.2.130/32 is directly connected, Vlan60
O 10.33.2.144/30 [110/2] via 10.33.2.153, 00:53:13, GigabitEthernet1/0/2
[110/2] via 10.33.0.1, 00:53:13, Vlan10
[110/2] via 10.33.0.129, 00:53:13, Vlan20
[110/2] via 10.33.1.1, 00:53:13, Vlan30
[110/2] via 10.33.1.129, 00:53:13, Vlan40
[110/2] via 10.33.2.1, 00:53:13, Vlan50
[110/2] via 10.33.2.129, 00:53:13, Vlan60
O 10.33.2.148/30 [110/2] via 10.33.2.157, 00:53:13, GigabitEthernet1/0/1
[110/2] via 10.33.0.1, 00:53:13, Vlan10
[110/2] via 10.33.0.129, 00:53:13, Vlan20
[110/2] via 10.33.1.1, 00:53:13, Vlan30
[110/2] via 10.33.1.129, 00:53:13, Vlan40
[110/2] via 10.33.2.1, 00:53:13, Vlan50
[110/2] via 10.33.2.129, 00:53:13, Vlan60
C 10.33.2.152/30 is directly connected, GigabitEthernet1/0/2
L 10.33.2.154/32 is directly connected, GigabitEthernet1/0/2
C 10.33.2.156/30 is directly connected, GigabitEthernet1/0/1
L 10.33.2.158/32 is directly connected, GigabitEthernet1/0/1
205.36.19.0/30 is subnetted, 4 subnets
O 205.36.19.0 [110/65] via 10.33.2.153, 00:53:13, GigabitEthernet1/0/2
O 205.36.19.4 [110/65] via 10.33.2.153, 00:53:13, GigabitEthernet1/0/2
O 205.36.19.8 [110/65] via 10.33.2.157, 09:04:01, GigabitEthernet1/0/1
Copy
Paste

7. Verifikasi VLAN (screenshot pada masing-masing multilayer switch)

MLS 1

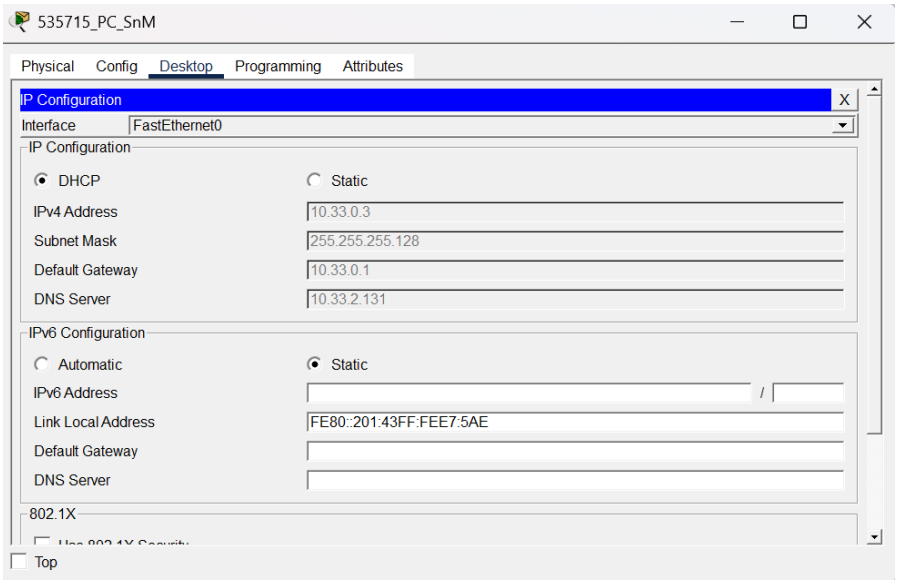


MLS 2

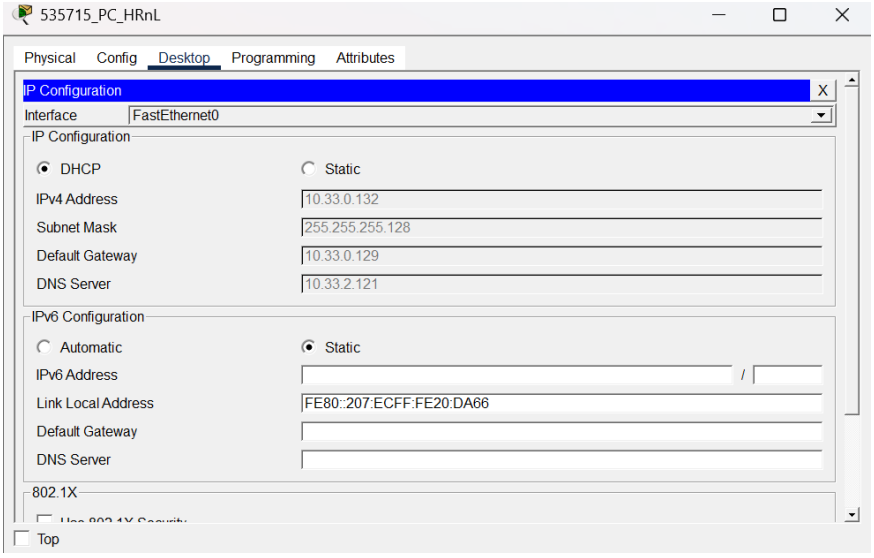


8. Verifikasi DHCP

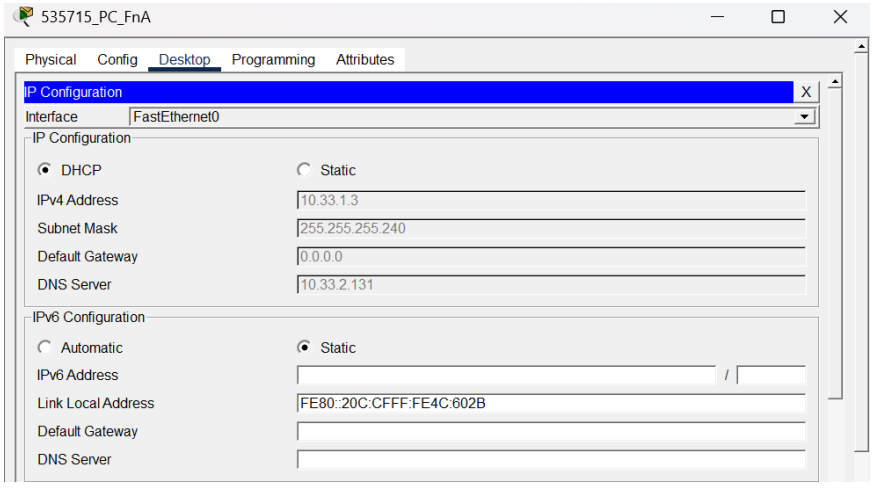
PC Sales and Marketing



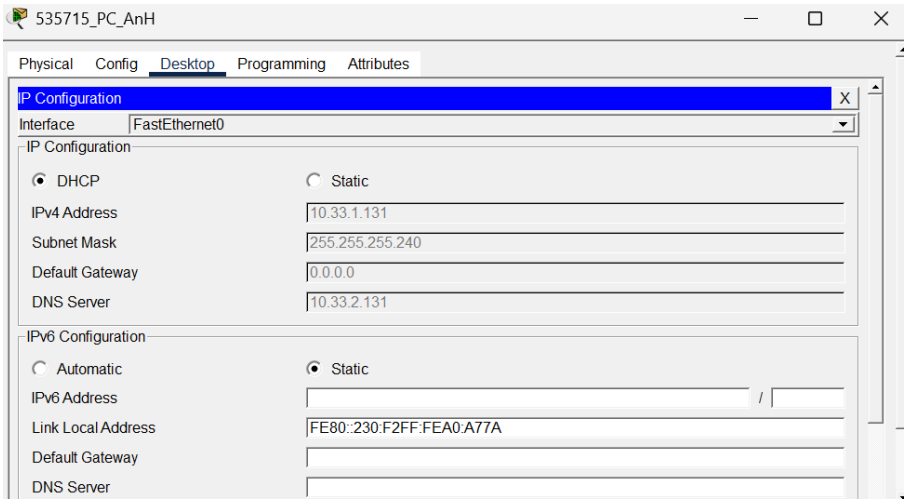
PC HR and Logistic



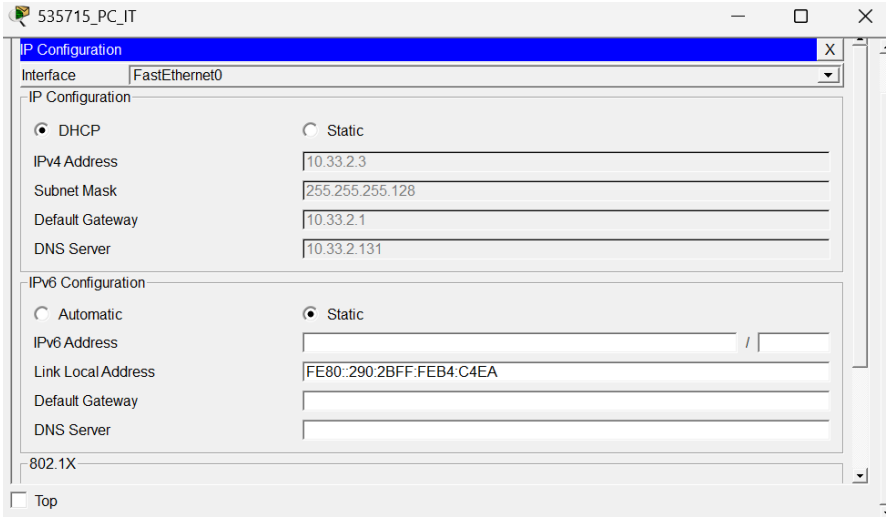
PC Finance and Administration



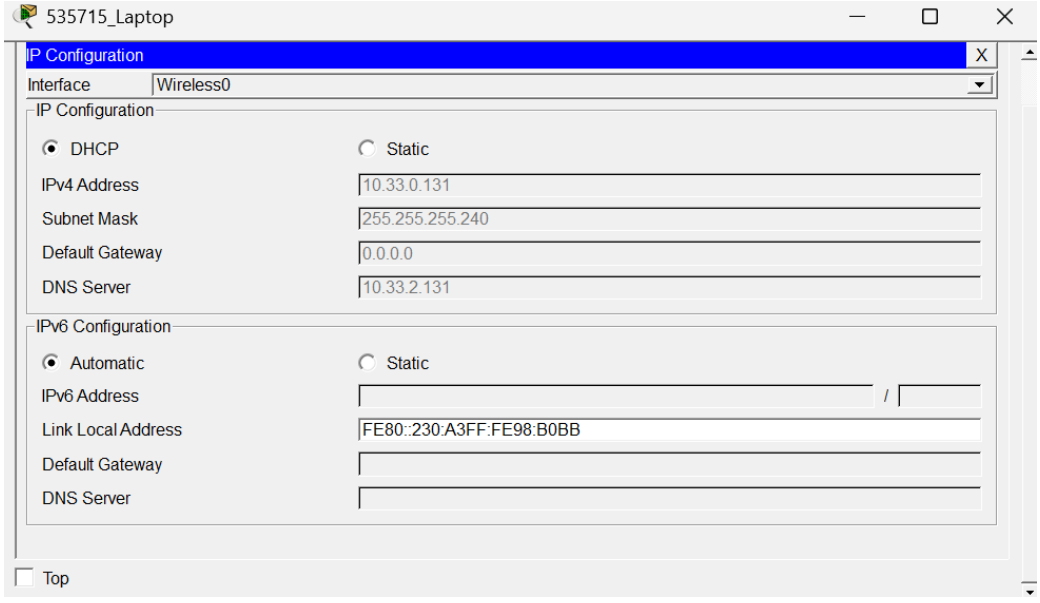
PC Admin and Humas



PC IT



Laptop



PEMBAHASAN

Praktikum ini difokuskan pada perancangan infrastruktur jaringan terintegrasi untuk gedung baru Perusahaan Nagoya. Struktur jaringan dibangun menggunakan arsitektur hierarkis dengan mengombinasikan dua multilayer switch dan dua router internal. Konektivitas luar didukung oleh jalur ganda ke dua ISP berbeda guna menjamin ketersediaan layanan yang berkelanjutan (redundancy).

Berdasarkan serangkaian pengujian, berikut adalah poin-poin utama dari hasil implementasi tersebut:

- Segmentasi dan Tata Kelola Trafik (VLAN & Inter-VLAN):
  Pemisahan lalu lintas data antar-departemen dilakukan secara logis melalui konfigurasi VLAN 10 hingga 60. Agar komunikasi antar-divisi tetap terjaga tanpa mengorbankan keamanan, fitur Inter-VLAN Routing diaktifkan pada multilayer switch sebagai pusat kendali aliran data.
- Otomatisasi Alamat IP (DHCP Helper):
  Meskipun layanan DHCP dikelola secara terpusat di ruang server, perangkat klien di setiap lantai tetap dapat memperoleh alamat IP secara otomatis. Hal ini dimungkinkan berkat penerapan IP helper-address pada multilayer switch yang bertindak sebagai perantara distribusi alamat IP ke setiap subnet departemen.
- Dinamika Rute dengan Protokol OSPF:
  Seluruh elemen jaringan, mulai dari multilayer switch hingga router ISP, dihubungkan menggunakan protokol routing dinamis OSPF. Melalui verifikasi perintah show ip route, sistem menunjukkan bahwa pertukaran informasi rute telah berjalan optimal, ditandai dengan munculnya kode 'O' pada tabel routing.
- Penguatan Keamanan Jaringan (SSH & Port-Security):
  Aspek keamanan manajemen perangkat diperketat dengan implementasi SSH versi 2 untuk mencegah penyadapan data saat akses jarak jauh. Selain itu, keamanan fisik pada akses switch diperkuat melalui fitur port-security dengan metode sticky dan violation shutdown, khususnya pada departemen Finansial dan Akunting untuk mencegah akses dari perangkat ilegal.

- Integrasi Koneksi Wireless dan Validasi Akhir:
  Penyediaan akses nirkabel dilakukan melalui pemasangan Access Point dengan standar keamanan WPA2-PSK di Departemen HR & Logistik. Seluruh rangkaian konfigurasi diakhiri dengan uji ping silang antar-departemen, di mana seluruh hasil pengujian menunjukkan konektivitas yang stabil dan bebas hambatan (no loss), baik pada media kabel maupun nirkabel..

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